

Responsible & Ethical Business Intelligence: Comprehensive Exam Reviewer

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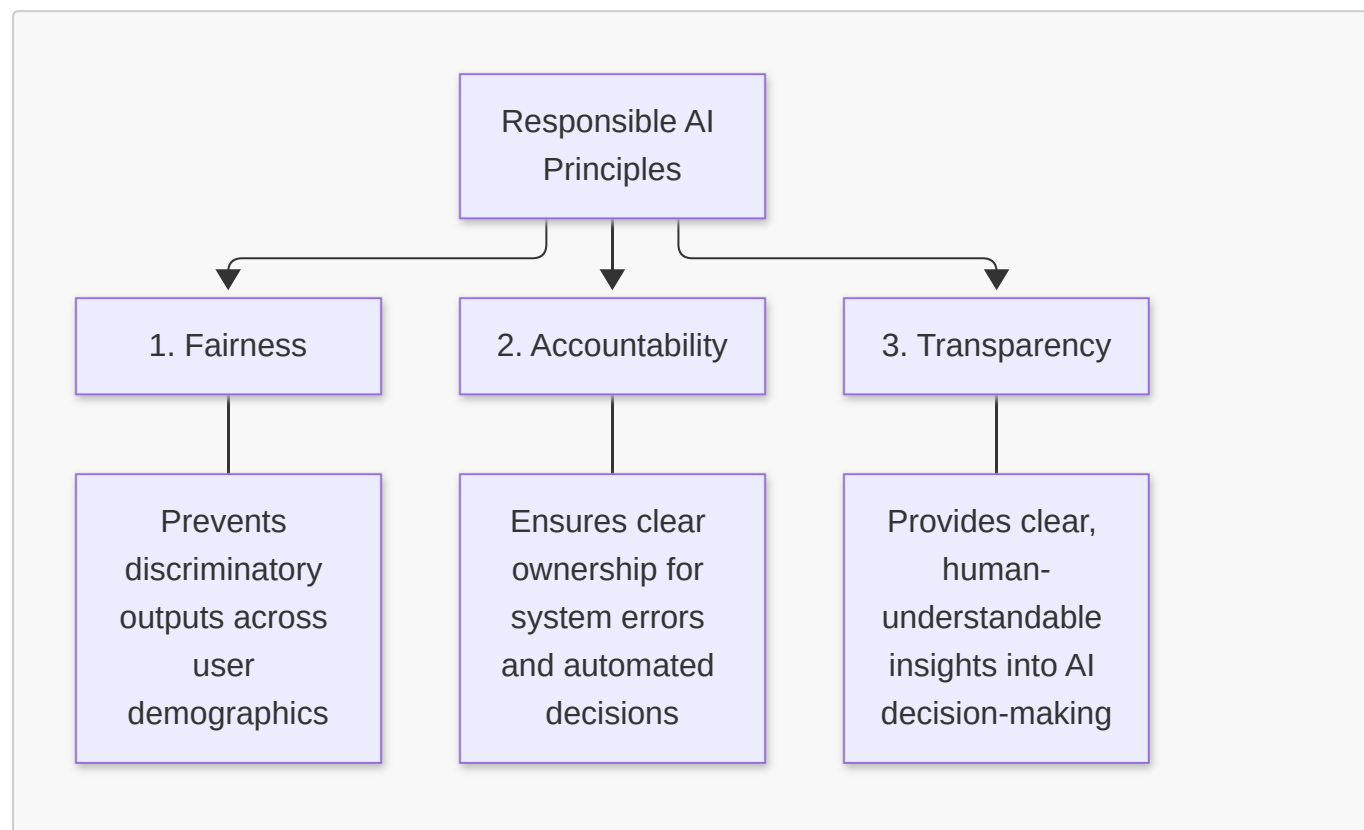
1. Core Concepts of Responsible AI

AI in Business Intelligence vs. Traditional BI

- **Traditional BI:** Relies heavily on human queries and manual analytical workflows to compile historical data.

- **AI-Powered BI:** Uses machine learning (ML) algorithms and automated analytics to process complex, unstructured datasets—revealing hidden patterns and automated predictive trends.
- **Responsible AI:** A governing framework of operational principles ensuring that AI deployment remains legal, ethical, transparent, and aligned with human values.

2. The 3 Pillars of Responsible AI

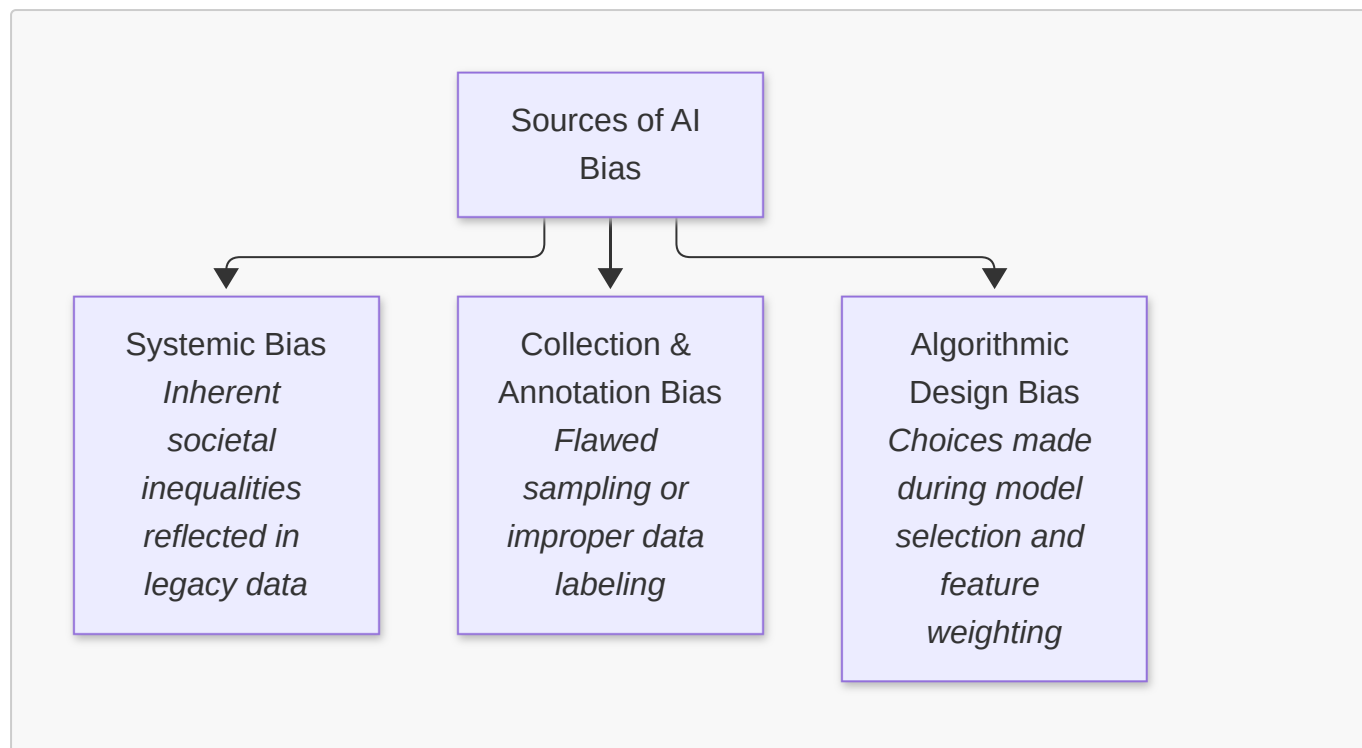


3. Fairness: Biases, Challenges & Mitigation

Why Fairness Matters

Guarantees equal opportunities across demographic groups (e.g., gender, ethnicity), enhances trust in enterprise BI tools, and prevents automated discrimination.

Sources of Bias in AI Systems



Mitigation Strategies for BI Teams

1. **Diverse Datasets**: Train algorithms on representative datasets matching target populations.
2. **Bias Detection Tools**: Deploy continuous auditing software to monitor disparate impact scores across groups.
3. **Regular Algorithmic Audits**: Evaluate outputs over time to detect emergent bias patterns.

4. Accountability: Black Boxes & Governance

Definition

Ensuring that organizations and developers are answerable for the outcomes, errors, and societal impacts of their deployed AI systems.

Primary Challenges to Accountability

- **The "Black Box" Problem**: Deep neural networks process parameters in ways that are difficult for humans to trace or comprehend.
- **Emergent Behavior**: Complex models can exhibit unintended actions that were never explicitly programmed.
- **Shared/Blurred Responsibility**: Difficult to assign singular blame when outputs involve data providers, developers, corporate users, and platform vendors.

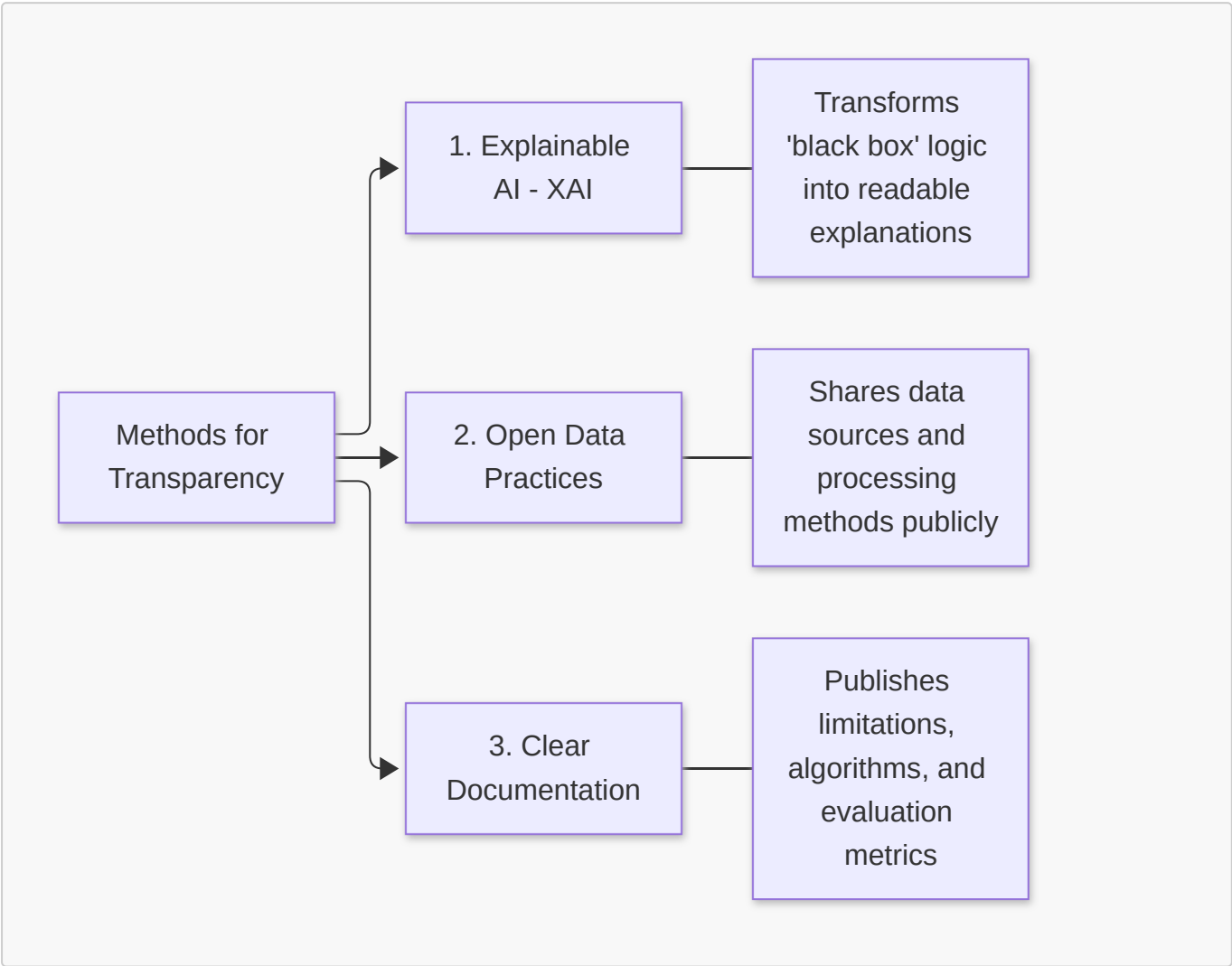
Strategies for Enforcing Accountability

- **Governance Frameworks**: Establish internal ethics boards, clear policies, and continuous risk management frameworks.
- **Traceable Documentation**: Maintain end-to-end documentation across the AI lifecycle (data provenance, training metrics, limitations).

- **Clear Ownership Lines:** Assign explicit authority and legal responsibility to individual roles and oversight bodies.

5. Transparency: Methods & Real-World Implementations

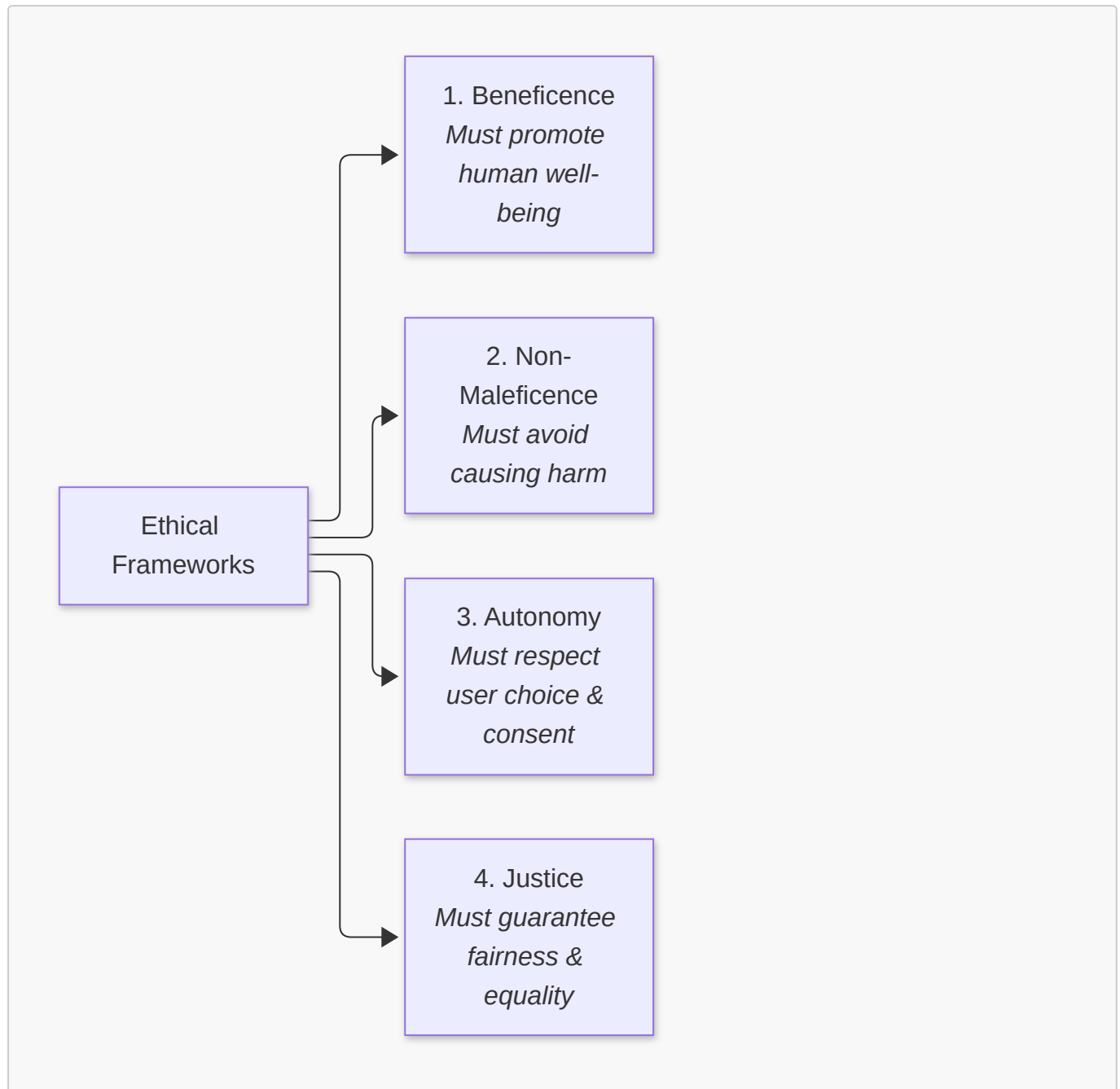
3 Core Methods to Achieve Transparency



Industry Implementations of AI Transparency

Company / Platform	Primary Focus	Implementation Example
Google	XAI & User Privacy	Introduced My Ad Center and search insights so users see why specific ads are targeted and adjust preferences.
IBM Watson	Healthcare Diagnostics	Breaks down medical literature, history, and imaging used to formulate a diagnostic recommendation for physicians.
Waymo	Autonomous Driving	Shares LiDAR/radar point-cloud visualizations, simulation datasets, and behavioral prediction models with regulators.
Meta (Facebook)	Feed Recommendations	Deployed the "Why am I seeing this?" feature to explain content algorithmic feeds and reduce misinformation.

6. Ethics & The 4 Core Principles



1. **Beneficence (Do Good):** Systems should be designed to improve human outcomes (e.g., early disease detection algorithms in healthcare).
2. **Non-Maleficence (Do No Harm):** Systems must prioritize safety and avoid injury (e.g., self-driving software prioritizing pedestrian safety).
3. **Autonomy (Respect Choice):** Users retain ultimate agency and right of refusal (e.g., customer service bots providing human-agent escalation options).
4. **Justice (Be Fair):** Eliminates discrimination across demographic lines (e.g., unbiased resume screening tools).

7. Real-World Case Studies of Unethical AI

Case Study 1: Algorithmic Discrimination (Law Enforcement)

- **Incident:** In 2020, **Robert Williams** (an African-American man) was wrongfully arrested in Detroit due to a false facial recognition match.
- **Technical Root Cause:** Training data lacked diverse representation. MIT studies found commercial facial recognition systems had an error rate of **34.7% for dark-skinned women vs 0.8% for light-skinned men**.
- **Violated Principles:** **Fairness, Justice, and Non-Maleficence.**

Case Study 2: Privacy Breach & Data Manipulation (Elections)

- **Incident:** The **Cambridge Analytica / Facebook Scandal (2018)** involved unauthorized harvesting of 87 million user profiles without explicit consent.
 - **Technical Root Cause:** Data was collected via a third-party quiz app ("This Is Your Digital Life") which scraped data from quiz takers and their friends list, feeding psychological profiling models for political manipulation.
 - **Regulatory Result:** Meta was fined **\$5 Billion by the U.S. FTC** and accelerated the global enforcement of strict privacy regulations like the EU GDPR.
 - **Violated Principles:** **Autonomy, Consent, and Data Privacy.**
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8. Philippine AI Legislative Landscape

The Philippines has introduced legislative measures to regulate AI development, labor protection, and ethics:

- **House Bill No. 7396:** *Artificial Intelligence Development and Regulation Act of the Philippines* (Focuses on national development and regulatory structures).
 - **House Bill No. 9448:** *Protection of Labor Against Artificial Intelligence Automation Act* (Protects workforce displacement and employee rights against unchecked automation).
 - **House Bill No. 7913:** *Artificial Intelligence Regulation Act* (Establishes ethical guidelines and compliance standards for AI systems).
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9. Quick-Recall Exam Prep & Cheat Sheet

High-Yield Facts & Key Distinctions

- **XAI (Explainable AI):** The specific technological domain designed to convert complex "black box" decisions into human-interpretable logic.
- **Emergent Behavior:** When an AI model develops unpredictable behaviors or outputs that developers did not explicitly code or intend.
- **Disparate Error Rates:** Facial recognition systems historically exhibit significantly higher error rates on dark-skinned individuals due to unbalanced training sets.
- **Consent vs. Usage:** **Consent** means getting explicit user permission *before* gathering data; **Data Usage** governs *how* that collected data is processed ethically without deceptive intent.
- **Key Philippine AI Laws:**
 - HB 7396 (Development & Regulation)
 - HB 9448 (Labor Protection)
 - HB 7913 (AI Regulation)